

Artificial Intelligence and Retrieval-Augmented Generation

Introduction

Artificial Intelligence (AI) is a field of computer science focused on building systems that can perform tasks that typically require human intelligence. These tasks include language understanding, image recognition, decision-making, and reasoning.

Large Language Models (LLMs) such as those developed by [OpenAI](#) are trained on massive amounts of text data. They can generate coherent responses, summarize documents, and answer questions. However, they have limitations.

One key limitation is that LLMs do not inherently know about documents that were created after their training cutoff or private documents that are not part of their training data.

Retrieval-Augmented Generation (RAG) addresses this limitation by allowing the model to search a document collection and use relevant passages when generating answers.

What Is Retrieval-Augmented Generation?

RAG is a framework that combines two components:

1. A retrieval system.
2. A generation model.

The retrieval system searches for relevant text chunks based on the user's query.

The generation model uses the retrieved chunks as context to produce an answer.

This approach improves factual accuracy and allows AI systems to answer questions grounded in specific documents.

Real-World Applications of RAG

RAG systems are used in many industries.

Healthcare

Doctors can query research papers and medical guidelines.

Legal

Lawyers can search contracts and case documents.

Education

Students can ask questions about textbooks and lecture notes.

Customer Support

Companies can build chatbots that answer questions from internal documentation.

Finance

Analysts can query annual reports and financial statements.

Why RAG Is Powerful

RAG provides several advantages:

- Reduces hallucinations.
- Uses private documents.
- Keeps responses grounded in source material.
- Supports citations.
- Updates knowledge without retraining.

For example, if a user uploads a company handbook, the system can answer policy questions directly from that handbook.

Document Chunking

What Is Chunking?

Chunking is the process of splitting a document into smaller sections.

Large documents may exceed the model's context window, so dividing them into chunks makes retrieval more efficient.

Common Chunk Sizes

- 200 characters
- 500 characters
- 1000 characters
- 1500 characters

Overlap

Overlap repeats some text between adjacent chunks.

Example:

- Chunk 1: sentences 1–5
- Chunk 2: sentences 4–8

Overlap preserves context across chunk boundaries.

Why Chunk Size Matters

If chunks are too small:

- Important context may be lost.

If chunks are too large:

- Irrelevant information may be retrieved.

Choosing the right chunk size is an optimization problem.

Embeddings

What Are Embeddings?

Embeddings are numerical vector representations of text.

Words and sentences with similar meanings have vectors that are close together in high-dimensional space.

For example:

- "dog" and "puppy" have similar embeddings.
 - "car" and "vehicle" have similar embeddings.
-

Embedding Models

Popular embedding models include:

- text-embedding-3-small by [OpenAI Embeddings Documentation](#)
 - BGE embeddings by [BAAI](#)
 - E5 embeddings by [Microsoft Research](#)
-

Similarity Search

Once embeddings are generated, similarity search is performed using metrics such as:

- Cosine similarity
- Euclidean distance
- Dot product

The most similar chunks are returned to the model.

Vector Databases

A vector database stores embeddings and supports efficient nearest-neighbor search.

Popular options include:

- [FAISS](#)
- [Chroma](#)

- [Pinecone](#)
- [Weaviate](#)

FAISS is widely used for local experimentation.

Query Flow in a RAG System

1. User uploads a PDF.
 2. PDF text is extracted.
 3. Text is split into chunks.
 4. Embeddings are generated.
 5. Embeddings are stored in a vector database.
 6. User asks a question.
 7. Query embedding is created.
 8. Similar chunks are retrieved.
 9. Retrieved chunks are sent to the LLM.
 10. The model generates the answer.
-

Example Questions

Your system should answer:

- What is RAG?
 - Why is chunk overlap important?
 - What are embeddings?
 - Name three vector databases.
 - What is cosine similarity?
 - How does FAISS work?
 - What industries use RAG?
-

Multi-Agent Systems

What Is an AI Agent?

An AI agent is a system that can perceive information, reason about it, and take actions.

An agent may:

- Search the web.

- Query databases.
 - Generate reports.
 - Call APIs.
-

Multi-Agent Architecture

In multi-agent systems, several specialized agents collaborate.

Examples:

- Research Agent
- Summarization Agent
- Coding Agent
- Review Agent

Each agent has a dedicated responsibility.

Benefits of Multi-Agent Systems

- Better modularity
 - Easier debugging
 - Role specialization
 - Parallel execution
 - Improved scalability
-

Time Series Forecasting

Time series data consists of observations recorded over time.

Examples:

- Stock prices
- Temperature
- Sales
- Website traffic

Popular forecasting models:

- ARIMA
- Prophet
- LSTM

- Transformer models
-

Prompt Engineering

Prompt engineering is the practice of designing effective instructions for language models.

Good Prompt Example

"Answer the question using only the provided context. If the answer is not in the context, say 'I don't know.'"

This reduces hallucination and improves trustworthiness.

Evaluation Metrics

RAG systems can be evaluated using:

- Precision
 - Recall
 - F1 Score
 - Faithfulness
 - Answer Relevance
 - Context Relevance
-

Sample Facts for Testing

- RAG stands for Retrieval-Augmented Generation.
 - FAISS was developed by Meta.
 - Embeddings convert text into vectors.
 - Cosine similarity measures angular closeness.
 - Chunk overlap helps preserve context.
 - Multi-agent systems use multiple specialized agents.
 - ARIMA is a classical time series model.
-

Conclusion

RAG is one of the most practical ways to connect language models with private or dynamic knowledge bases.

By understanding document chunking, embeddings, vector databases, retrieval, and prompting, developers can build accurate and explainable AI applications.

Your PDF Q&A project is an excellent hands-on introduction to modern AI engineering. It combines natural language processing, information retrieval, and software development into one practical system.